

E-Commerce Return Intelligence & Customer Risk Analytics

Project Documentation

1. Project Objective

The main objective of this project was to understand e-commerce return patterns and identify areas where return rates were higher. I worked on the project from data cleaning through analysis, SQL, Power BI, and business recommendations.

2. Business Problem

Product returns can affect revenue, inventory planning, customer experience, and business operations. I wanted to understand which product categories, price levels, customer characteristics, and order-related factors were associated with different return rates.

The analysis looks at observed patterns and associations. It does not assume that a particular factor directly causes a return.

3. Dataset

The dataset contains 200,000 e-commerce orders. It includes customer, product, browsing, delivery, payment, shipping, coupon, and return information.

- Customer age and previous purchase history
- Previous return rate
- Product price, discount, and product rating
- Session length and product views
- Device type and product category
- Shipping and payment methods
- Coupon usage and delivery delay
- Return status

4. Data Cleaning

During the data-quality check, I found some values that were not logically valid for their respective columns. Instead of changing these values to zero, I converted them to NaN so they would be treated as missing values during the analysis.

- Negative product prices
- Negative discount percentages
- Negative session lengths
- Negative product views
- Negative past return rates
- Product ratings below 1 or above 5

I also found some customer ages below 18. I kept these values because there was not enough information to conclude that they were invalid.

5. Exploratory Data Analysis

I used exploratory analysis to compare return behavior across product categories, customer characteristics, product price, browsing behavior, delivery, and other order-related factors.

Overall Return Rate

Metric	Value
Total orders	200,000
Returned orders	94,919
Not returned orders	105,081
Overall return rate	47.46%

Product Category

The observed return rates varied across product categories. Clothing had the highest observed return rate, while Electronics had the lowest.

Product Category	Return Rate
Clothing	53.05%
Toys	50.31%
Beauty	49.72%
Sports	45.04%
Home	44.54%
Electronics	41.73%

Customer Experience

- Low experience: 50.68%
- Medium experience: 45.61%
- High experience: 47.16%

The groups showed different observed return rates, indicating an association between customer experience level and return behavior.

Product Price

- Low price: 46.20%
- Medium price: 47.86%
- High price: 50.24%

6. Feature Engineering

I created additional groups so that different parts of the dataset could be compared more easily.

- Age Group
- Return History Group
- Delivery Time
- Discount Group
- Price Group

- Session Length Group
- Product View Group
- Customer Experience Group

7. Return-Risk Segmentation

For the main segmentation analysis, I combined product category, price group, and customer experience group. This produced 54 segments.

The overall return-rate baseline was 47.46%. Each segment was compared with this baseline to identify segments with higher or lower observed return rates.

Highest observed segment: Clothing + High Price + Low Customer Experience

- Orders: 1,628
- Return rate: 60.57%
- Gap from baseline: +13.11 percentage points

Segment classification used in the analysis:

- Elevated: at least 5 percentage points above the baseline
- Near baseline: within 5 percentage points of the baseline
- Lower: at least 5 percentage points below the baseline

The ± 5 percentage-point value was used as a practical threshold for this project.

8. SQL Analysis

I used MySQL to perform business-focused analysis on the dataset. The analysis covered order counts, return rates, categories, shipping, payment, coupon usage, delivery conditions, product prices, and customer purchase history.

The SQL queries are available in `sql/ecommerce_analysis.sql` in the repository.

9. Power BI Dashboard

The Power BI dashboard was divided into three pages so the analysis could be viewed from an overall level and then explored in more detail.

Page 1 – Executive Overview

- Total orders
- Returned and not returned orders
- Overall return rate
- Product category return rates
- Delivery conditions

Page 2 – Customer & Product Analysis

- Customer experience
- Price
- Session length
- Payment method
- Discount level
- Product views

- Shipping method
- Coupon usage

A matrix was also used to compare price group and customer experience.

Page 3 – Return Risk Segmentation

- Product category
- Price group
- Customer experience
- Return rate
- Overall baseline
- Risk gap
- Highest-risk segment

10. Key Findings

- Clothing had the highest observed category return rate at 53.05%.
- Electronics had the lowest observed category return rate at 41.73%.
- Higher price groups showed higher observed return rates.
- Customer experience groups showed different observed return rates.
- Combining category, price, and customer experience revealed stronger differences between segments.
- Clothing + High Price + Low Customer Experience had a 60.57% observed return rate, 13.11 percentage points above the overall baseline.

11. Business Recommendations

1. Focus on higher-return clothing segments, especially higher-priced clothing with lower customer experience.
2. Improve product information for higher-priced products.
3. Investigate customer experience patterns associated with higher returns.
4. Review clothing-specific product information and customer expectations.
5. Monitor delivery patterns and investigate the difference between delivery conditions.
6. Improve product information for customers with fewer product views.
7. Review whether shorter sessions indicate insufficient product information.
8. Treat discounts as a lower-priority factor because the observed difference between discount groups was very small.
9. Prioritize elevated return-risk segments instead of treating all customers and products equally.
10. Use lower-return segments as comparison benchmarks.
11. Focus business attention on category, price, customer experience, and their combinations.
12. Monitor these segments over time to see whether the patterns remain consistent.

These recommendations are based on observed patterns in the dataset and should be validated with additional business or operational data before major decisions are made.

12. Conclusion

This project helped me work through an end-to-end data analytics workflow, starting with data cleaning and exploratory analysis and continuing through feature engineering, risk segmentation, SQL analysis, Power BI visualization, and business recommendations.

The main focus was to identify useful patterns in e-commerce returns and understand where further business investigation could be focused.